Mandatory deliverable

LAB | Your First AWS Lambda Function Lab

Introduction

In this hands-on lab, you will explore **AWS Lambda**, a serverless computing service that runs code in response to events. This lab will provide practical experience in creating, deploying, and testing Lambda functions.

Learning Objectives

By completing this lab, you will be able to:

1. **Create an AWS Lambda function** using the AWS Management Console.
2. **Understand key AWS Lambda components**, including event sources, execution roles, and runtime environments.
3. **Invoke the function manually** and using an event trigger.
4. **Monitor function execution** using Amazon CloudWatch logs.
5. **Explore AWS Lambda use cases** and best practices.

Lab Instructions

1. Create an AWS Lambda Function

1. Navigate to **AWS Management Console** → **Lambda Service**.
2. Click **Create Function**.
3. Configure the function:
   - **Function Name**: `lab-lambda-function-<your-name>`
   - **Runtime**: Choose Python 3.x, Node.js, or another preferred runtime.
   - **Execution Role**: Create a new role with **basic Lambda execution permissions**.
   - **Code**: Use the built-in editor or upload a .zip package.
4. Click **Create Function**.

2. Write and Deploy Your Lambda Function

Ask AI Tutor

1. In the function code editor, enter the following Python example:

```
def lambda_handler(event, context):
    return {"statusCode": 200, "body": "Hello from AWS Lambda!"}
```

Copy Explain

2. Click **Deploy** to save the function.

3. Invoke Your Lambda Function

1. In the **Lambda Console**, navigate to the **Test** tab.
2. Click **Create Test Event**, name it `test-event`, and use the default JSON payload.
3. Click **Invoke** and verify the response.

4. Set Up an Event Trigger

1. Go to the **Configuration** tab and select **Triggers**.
2. Click **Add Trigger**.
3. Choose **API Gateway** as the event source (You could optionally try other triggers like Amazon S3, DynamoDB, or CloudWatch).
4. Configure the trigger settings and save changes.

5. Monitor Function Execution

1. Navigate to **Amazon CloudWatch** → **Logs**.
2. Locate the log group for your Lambda function.
3. View execution logs and analyze function performance.

6. (Optional) Explore Lambda Function Options

1. Adjust **memory allocation and timeout settings** under the **Configuration** tab (you'd need this for Python pandas excel document manipulation).
2. Enable **provisioned concurrency** for better performance.
3. Explore **AWS X-Ray** for tracing function execution.

Completion Criteria

- **AWS Lambda function created** with correct configurations.
- **Successful manual invocation** of the function.
- **Event trigger set up and tested** successfully.
- **CloudWatch logs reviewed** for monitoring execution.

Submission Guidelines

- Take **screenshots** of key steps and results.
- Compile them into a **Google Doc** and upload them to **Google Drive**.
- Share the **Google Drive link** with the instructor and ensure they have access.

🚀 **Congratulations! You've successfully created and deployed an AWS Lambda function!** 🎉

Lab completed!

Submitted on Sept 23, 2026 a las 15:37

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Progress of Week 2 **6/43**

You have 37 units left to complete

Mandatory deliverables **6/11**